

SANDEEP REDDY REDDY

Senior Software Engineer | Full Stack & Cloud Engineering | AI / Security / GRC

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PROFESSIONAL SUMMARY

Senior Software Engineer and 2x Patent Inventor with 8+ years building production-grade Generative AI systems, distributed platforms, and enterprise security automation at IBM. Proven architect of multi-agent LLM frameworks, DevSecOps pipelines, and GRC platforms serving IBM's global workforce — with a track record of shipping high-impact systems end-to-end from design through production at enterprise scale.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, Bash, SQL

AI / GenAI: LLMs (Open-Source & Proprietary), Multi-Agent Systems, RAG, Prompt Engineering, Agentic AI, watsonx Assistant, watsonx.ai, watsonx Orchestrate, Watson Discovery

Backend: Spring Boot, Node.js, FastAPI, Flask, REST APIs, GraphQL, gRPC, Microservices, Event-Driven Architecture, Java 8 (Streams, Lambda)

Frontend: React, Next.js, Angular, TypeScript, HTML5, CSS3, Bootstrap

Cloud & DevOps: IBM Cloud, Docker, OpenShift, Kubernetes, Serverless, CI/CD (Jenkins, GitHub Actions), Infrastructure as Code, NGINX, Container Registry

Security: OWASP Top 10, DevSecOps, SAST, DAST, Software Supply Chain Security (SBOM), Secrets Detection (Gitleaks), Zero Trust, Vulnerability Management

GRC & Risk: Third-Party Risk Management (TPRM), GRC (ProcessUnity), Vendor Risk Assessment, SOX, PCI-DSS, FFIEC, SSO/SAML, Secure SDLC

Databases: NoSQL (Cloudant, MongoDB), SQL (PostgreSQL, MySQL), Object Storage (S3/B2), Redis / In-Memory Caching, CycloneDX/SPDX

Tools & Testing: SonarQube, JUnit, Postman, Swagger/OpenAPI, Maven, Gradle, Git, Slack SDK, ZenHub

PROFESSIONAL EXPERIENCE

Senior Software Engineer — IBM

May 2021 – Present

Third-Party Security Risk Management (TPSRM) | Durham, NC

- Engineered a production Generative AI conversational assistant (watsonx Assistant, FastAPI) deployed on Slack and web — serving IBM's global enterprise workforce across security operations; automated 24/7 vendor risk status inquiries, eliminated all routine support tickets, and delivered real-time risk alert notifications; reduced average response time from hours to under 10 seconds.
- Architected a fully autonomous multi-agent AI system (watsonx Orchestrate) with specialized agents for discovery, evaluation, and reporting — auto-ingesting vendor security posture from public trust centers and artifacts; achieved 100% end-to-end process automation, eliminated all manual supplier engagement, and compressed assessment cycle time by 7+ days.
- Designed a dual-LLM validation engine (watsonx.ai + LLaMA 3-70B) to score supplier service descriptions in real time — improving submission quality 77%, cutting manual review effort 70%, and achieving 94% AI accuracy across thousands of assessments validated via double-blind testing.
- Engineered an end-to-end workflow automation platform (ProcessUnity, Monday.com, Mend) for SBOM assessment orchestration — slashing setup time 90%+ (30+ min → 2–3 min) and eliminating manual data entry from the vendor risk pipeline.
- Built a software supply chain security scanning pipeline (Mend/WhiteSource) — achieving 100% SBOM dependency scan coverage for critical/high-severity CVEs across all supplier software, replacing manual research with systematic automated detection.
- Designed and deployed a multi-stage DevSecOps CI/CD pipeline (Jenkins, GitHub Actions) with SAST, container scanning, dependency analysis, and secrets detection (Gitleaks) — cutting deployment time from hours to minutes and increasing release cadence 4x (monthly → weekly).
- Architected and administered IBM Cloud infrastructure for the TPSRM platform — provisioning Code Engine (serverless), Container Registry, Cloud Object Storage, Secrets Manager, and IBM Logs; configured IAM policies and AccessHub for team

onboarding and CBN elevated-role governance; deployed watsonx, Watson Discovery, Watson Studio, and CloudantDB at enterprise scale.

- Enforced OWASP Top 10 secure coding standards across all TPSRM services; led vulnerability assessments using IBM App Scan and Contrast Security, remediating XSS, SQL injection, and injection flaws across production; administered ProcessUnity GRC platform for enterprise-wide governance, risk, and compliance workflow management.
- Engineered SSO/SAML identity federation and directory integration (IBM W3id, Bluepages) — unifying authentication and enabling secure single sign-on across 5+ internal tools and third-party SaaS applications, eliminating siloed identity management.
- Architected automated integrations for SIG Questionnaire and Black Kite risk intelligence via custom REST APIs and ProcessUnity templates — enabling real-time external risk signal ingestion and driving the GRC platform roadmap for SOX, PCI-DSS, and FFIEC compliance.
- Built a unified data aggregation layer to consolidate risk datasets across IBM Procurement, Risk Management, Security, and Business Units — delivering a single source of truth for enterprise risk reporting and audit-ready compliance pipelines.
- Technical lead across 6+ distributed TPSRM services serving IBM's global enterprise — owned system design and architecture decisions end-to-end; led code and design reviews, defined team engineering standards, mentored junior engineers, and partnered cross-functionally with Security, Product, Legal, and Compliance leadership to ship reliable, compliant systems at scale.

Software Engineer — IBM (via Inteliroute Technologies LLC, Contractor)

Apr 2018 – May 2021

Agile Risk Management → Third-Party Risk Management | Durham, NC | Client: IBM

- Built and maintained the Agile Risk Management (ARM) platform at IBM — a deviation risk tracking system enabling project teams to identify, assess, and escalate schedule and delivery risks; engineered full-stack features (Java Spring Boot, Angular, REST APIs) with end-to-end ownership across multiple release cycles.
- Transitioned to IBM's Third-Party Risk Management (TPRM) program — contributed to the foundational buildout of ProcessUnity GRC workflows, SSO/SAML identity integrations, and microservice-based vendor assessment pipelines that later scaled into the enterprise TPSRM platform.
- Designed and implemented SSO/SAML federated identity (IBM Liberty + ProcessUnity SaaS) and integrated SendGrid for enterprise transactional email delivery — establishing secure, scalable identity and notification infrastructure for the TPRM program.
- Designed NoSQL (Cloudant) data integration workflows and RESTful/SOAP microservices for vendor assessment pipelines; authored full Swagger/OpenAPI documentation and Postman test suites ensuring API reliability across all endpoints.
- Containerized microservices with Docker, deployed on IBM Hybrid Cloud (OpenShift), configured NGINX for load balancing, and built Jenkins CI/CD pipelines — enabling fully automated, repeatable build and deployment workflows.
- Developed reusable Angular component libraries and applied Java 8 (Streams, Lambda) and Python (Pandas, Beautiful Soup) for backend data processing, automated vendor data enrichment, and scalable API integrations.
- Led end-to-end system testing, UAT coordination, and production incident resolution — maintaining platform reliability, uptime, and enforcing security and compliance standards across all release cycles.

PATENTS

- **Detection and Management of Anomalies in Assessment Data**

U.S. Patent Application No. 19/008667 • IBM • Filed Jan 2025

Covers automated anomaly detection and management within third-party security assessment data pipelines to improve data integrity and risk accuracy.

- **System and Method for Automated Verification of Supplier Risk Questionnaire Data Using Multi-Model Language Understanding**

IBM Invention Disclosure P202501766 • Filed May 2025 • Review: Completed

Covers multi-model LLM-based automated verification of supplier risk questionnaire submissions — directly tied to the dual-LLM validation engine built for the TPSRM platform.

EDUCATION

Master of Science in Computer Science

Dec 2017

Texas A&M University, TX, USA

Bachelor of Engineering in Computer Science

Apr 2016

SRM University, India